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Saveetha Institute of Medical And Technical Sciences
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**INTELLIGENT EMPLOYEE ATTENDANCE, PAYROLL, AND
PERFORMANCE MANAGEMENT SYSTEM USING ADVANCED
OBJECT-ORIENTED PROGRAMMING CONCEPTS IN C++
COMPARED WITH TRADITIONAL EMPLOYEE MANAGEMENT
SYSTEMS FOR IMPROVED WORKFORCE ADMINISTRATION AND
PAYROLL ACCURACY**

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

DSA0112 OBJECT ORIENTED PROGRAMMING WITH C++
to the award of the degree of

**BACHELOR OF ENGINEERING IN
COMPUTER SCIENCE ENGINEERING (ARTIFICIAL INTELLIGENCE)**

Submitted by

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Under the Supervision of

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BONAFIDE CERTIFICATE

This Is To Certify That The Capstone Project Entitled “**Intelligent Employee Attendance, Payroll, And Performance Management System Using Advanced Object-Oriented Programming Concepts In C++ Compared With Traditional Employee Management Systems For Improved Workforce Administration And Payroll Accuracy**” Has Been Carried Out By **K. Sri Lasya (192472216)** Under The Supervision Of **Dr. I. Sudha** And Is Submitted In Partial Fulfilment Of The Requirements For The Current Semester Of The **B.E.,CSE(AI)** Program At Saveetha Institute Of Medical And Technical Sciences, Chennai.

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DECLARATION

We, **K. Sri Lasya** of the **CSE(AI)**, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled **“Intelligent Employee Attendance, Payroll, And Performance Management System Using Advanced Object-Oriented Programming Concepts In C++ Compared With Traditional Employee Management Systems For Improved Workforce Administration And Payroll Accuracy”** is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Signature of the Students with Names

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Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
K. Sri Lasya	Employee information and attendance management	Developed employee registration, employee records, attendance and leave management using classes, objects and encapsulation	Tested employee data entry, record retrieval, attendance and leave functions	Introduction, Problem Statement, Objectives, Scope	35%
Lakshmi Devi	Payroll calculation and salary management	Developed salary calculation, attendance-based deductions and payroll processing using OOP concepts	Tested salary calculations, deductions and different attendance cases	Literature Review, Payroll Methodology, Functional Requirements	35%
Anu Keerthana	Performance evaluation, reporting and system integration	Developed performance evaluation, ratings, report generation, file handling and integrated system functionality	Tested performance ratings, file storage/retrieval, reports and overall system integration	System Design, Testing & Results, Conclusion, Future Scope, References	30%
Total					100%

ABSTRACT

Employee management is an essential part of every organization, involving the maintenance of employee information, attendance tracking, payroll processing, and performance evaluation. Traditional employee management methods often depend on manual record-keeping and separate processes, which can be time-consuming, difficult to maintain, and prone to human errors. The proposed **Intelligent Employee Attendance, Payroll, and Performance Management System** addresses these limitations by integrating major employee-management activities into a single C++-based application.

The system is designed using **Advanced Object-Oriented Programming (OOP) concepts**, including **Classes, Objects, Encapsulation, Inheritance, Polymorphism, and File Handling**. These concepts provide a structured and modular approach for developing the application and help improve code organization, reusability, security, and maintainability. C++ is used to implement the system because it provides strong support for object-oriented design and structured data management.

The proposed system consists of three major modules: **Employee Information and Attendance Management, Payroll Calculation and Salary Management, and Employee Performance Evaluation and Reporting**. The employee management module stores important details such as Employee ID, name, department, designation, and contact information. The attendance module records daily attendance and maintains employee status such as Present, Absent, or Leave. The collected attendance information is then used by the payroll module for salary processing and attendance-based deductions. The performance module evaluates employees using performance scores, productivity, and attendance and generates suitable performance reports. These integrated functions are directly specified in the project PPT.

File handling is used to store and retrieve employee information, allowing records to be maintained for future use. The system reduces repetitive manual calculations and provides faster access to employee records and reports. An object-oriented employee/payroll design is also a common practical application of C++ OOP principles.

The developed system successfully integrates employee information, attendance, payroll, and performance management into one application. Compared with traditional employee-management approaches, it provides **faster processing, better data organization, reduced manual effort, improved payroll accuracy, and reliable report generation**.

Keywords: Employee Management, Attendance Management, Payroll Management, Performance Evaluation, C++, Object-Oriented Programming, Encapsulation, Inheritance, Polymorphism, File Handling.

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CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.Introduction

Employee management is an important organizational process involving employee records, attendance, salary processing, and performance evaluation.

Traditional employee management systems commonly rely on manual records. Such systems require considerable administrative effort and can result in errors during attendance recording and payroll calculations.

The proposed system automates these activities using C++ and advanced Object-Oriented Programming concepts. The system provides an integrated platform for managing employee information, attendance, payroll, and performance.

The original project description identifies manual employee management, attendance tracking, payroll calculation, and separate records as the major problems addressed by the system.

1.2 Problem Statement

In an organization, HR personnel must maintain several types of employee information:

- Personal information
- Department details
- Designation
- Attendance
- Salary
- Performance
- Leave information

When these records are maintained manually, retrieving and updating information becomes difficult.

An integrated computerized system can connect these activities and provide faster access to employee information.

1.3 Aim

Traditional employee management systems depend heavily on manual record keeping.

The major problems are:

1. Manual employee record maintenance.
2. Time-consuming attendance tracking.
3. Human errors in payroll calculations.
4. Separate employee, attendance, payroll, and performance records.
5. Difficulty in retrieving employee information.
6. Increased administrative workload.
7. Lack of integrated reporting.

Therefore, an integrated employee management system is required to automate these processes and improve workforce administration

1.4 Objectives

The objectives of the project are:

- To develop an integrated employee management system.
- To maintain employee information efficiently.
- To automate attendance management.
- To automate payroll calculation.
- To reduce salary calculation errors.
- To evaluate employee performance.
- To generate employee reports.
- To implement advanced OOP concepts in C++.
- To use file handling for persistent data storage.

1.5 Scope of the Project

The scope of the project includes:

- Employee registration
- Employee record management
- Attendance recording
- Leave management
- Payroll calculation
- Salary deduction
- Performance evaluation
- Performance reporting
- File-based data storage
- Employee information retrieval

1.6 Proposed System

The existing traditional system generally uses:

- Paper records
- Spreadsheets
- Manual attendance registers
- Manual salary calculations
- Separate performance records

Disadvantages

- High manual effort
- Increased possibility of errors
- Slow data retrieval
- Data duplication
- Difficult report generation
- Poor integration

1.7 Proposed System

The proposed system integrates all major employee-management functions into one C++ application.



Fig. No. 1 Proposed Employee Management System

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Literature Survey

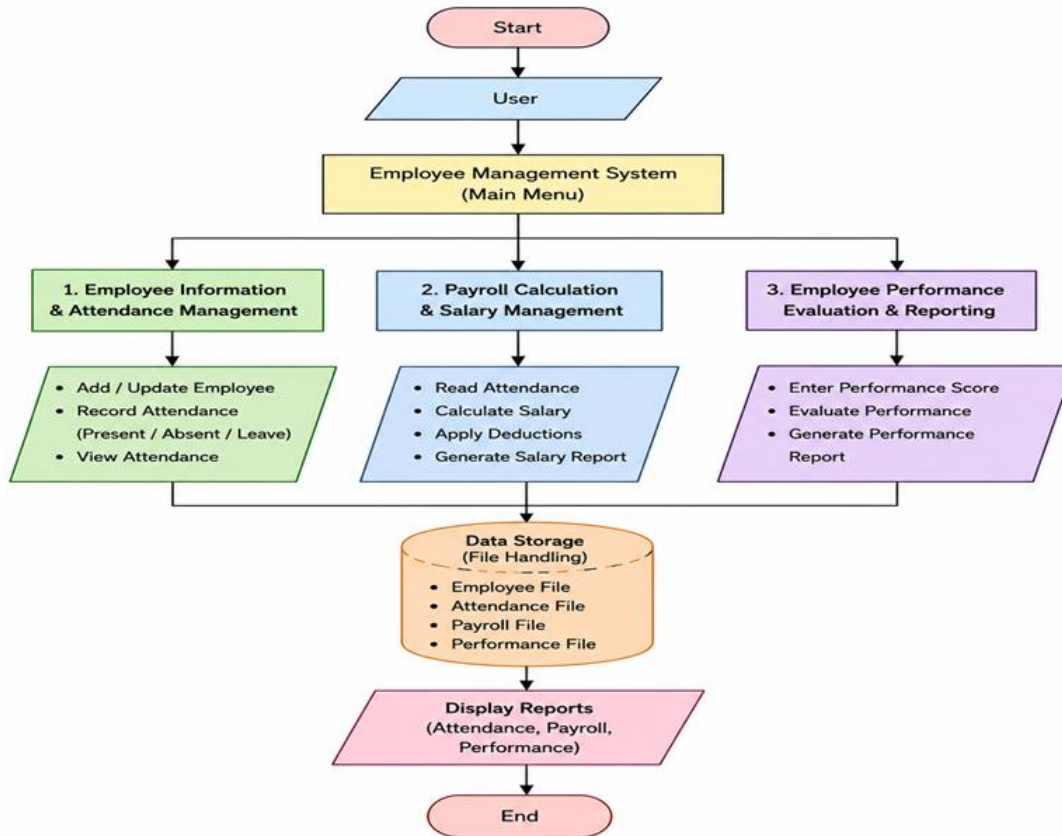
The literature survey in your PPT covers employee management, automated attendance, payroll, RFID/biometric attendance, and web-based payroll systems.

S.No.	Title	Authors	Year	Technique
1	Human Resource Management System	Syed Navaz et al.	2013	HRMS
2	Automated Attendance System Based on NFC and XBee	Yeboah-Boateng et al.	2016	NFC, XBee, Database
3	Payroll Management and Attendance Monitoring	Shukla & Bhandari	2019	RFID, Biometric
4	Employee Attendance and Payroll System	Sontakke et al.	2021	Attendance, Payroll
5	Web-Based Payroll Management System	Ahmed et al.	2023	Web System, Database

CHAPTER 3

SYSTEM DESIGN AND PROPOSED METHODOLOGY

3.1 System Architecture



3.2 Module 1 – Employee Information and Attendance Management

This module maintains employee details such as:

- Employee ID
- Name
- Department
- Designation
- Contact information

It also records daily attendance as:

- Present
- Absent
- Leave

The PPT specifically states that Classes, Objects, Encapsulation, and File Handling are used to organize and store employee information.

3.3 Module 2 – Payroll Calculation and Salary Management

The payroll module calculates employee salaries based on:

- Basic salary
- Working days
- Attendance
- Deductions

Basic formula

$$\text{Daily Salary} = \frac{\text{Basic Salary}}{\text{Total Working Days}}$$
$$\text{Attendance Deduction} = \text{Absent Days} \times \text{Daily Salary}$$
$$\text{Final Salary} = \text{Basic Salary} - \text{Attendance Deduction}$$

The actual payroll rules can be modified according to organizational requirements.

The PPT specifies that attendance-based deductions are applied and final salary is calculated automatically.

3.4 Sustainability

Employee performance is evaluated using:

- Performance score
- Productivity
- Attendance

The system assigns a performance rating based on predefined criteria.

Score	Rating
90–100	Excellent
75–89	Very Good
60–74	Good
50–59	Satisfactory
Below 50	Needs Improvement

3.5 OOPS CONCEPTS USED

OOP Concept	Application
Class	Employee, Payroll, Attendance, Performance
Object	Individual employee records
Encapsulation	Protecting employee data
Inheritance	Reusing employee properties

Polymorphism	Different employee behaviours
Abstraction	Hiding implementation details
File Handling	Storing employee records

3.6 Functional Requirements

The system should be able to:

1. Add employee information.
2. Display employee records.
3. Record attendance.
4. Calculate salary.
5. Apply deductions.
6. Evaluate performance.
7. Generate reports.
8. Save employee information.
9. Retrieve stored information.

3.7 Non-Functional Requirements

Performance

The system should process employee information quickly.

Reliability

Stored employee records should be retrievable accurately.

Maintainability

The use of OOP should make the program modular and easier to maintain.

Usability

The system should provide a simple interface for employee-management operations.

Security

Employee information should be protected through controlled access to data.

CHAPTER 4

IMPLEMENTATION, TESTING AND RESULTS

4.1 Implementation Environment

Hardware

- Computer/Laptop
- Minimum 4 GB RAM
- Keyboard
- Monitor

Software

- C++
- C++ Compiler
- Code::Blocks / Visual Studio / Dev-C++
- Windows/Linux operating system

4.2 Employee Registration

The administrator enters

Employee ID

Employee Name

Department

Designation

Contact Number

Basic Salary

4.3 Limitations

Example

Employee ID	Present	Absent	Leave
101	24	2	1
102	26	1	0
103	22	3	2

4.4 Payroll Calculation

Employee	Basic Salary	Working Days	Absent Days	Deduction	Final Salary
Employee A	₹30,000	26	2	₹2,307.69	₹27,692.31
Employee B	₹35,000	26	1	₹1,346.15	₹33,653.85

4.5 Performance Evaluation

Example

Employee	Score	Productivity	Attendance	Rating
Employee A	90	High	Good	Excellent
Employee B	78	High	Good	Very Good
Employee C	65	Medium	Average	Good

4.6 Testing

Test Case	Input	Expected Result	Status
TC01	Add employee	Employee stored	Pass
TC02	Display employee	Employee displayed	Pass
TC03	Mark attendance	Attendance updated	Pass
TC04	Calculate payroll	Correct salary generated	Pass
TC05	Apply deduction	Deduction calculated	Pass
TC06	Evaluate performance	Rating generated	Pass
TC07	Save record	Record stored in file	Pass
TC08	Retrieve record	Record displayed	Pass

4.7 COMPARISON WITH TRADITIONAL SYSTEM

Parameter	Traditional System	Proposed System
Employee Records	Manual	Automated
Attendance	Manual register	Computerized
Payroll	Manual	Automated
Performance	Separate records	Integrated
Data Retrieval	Slow	Fast
Calculations	Manual	Automatic
Error Possibility	High	Reduced
Reports	Manual	Automated
Data Organization	Low	High

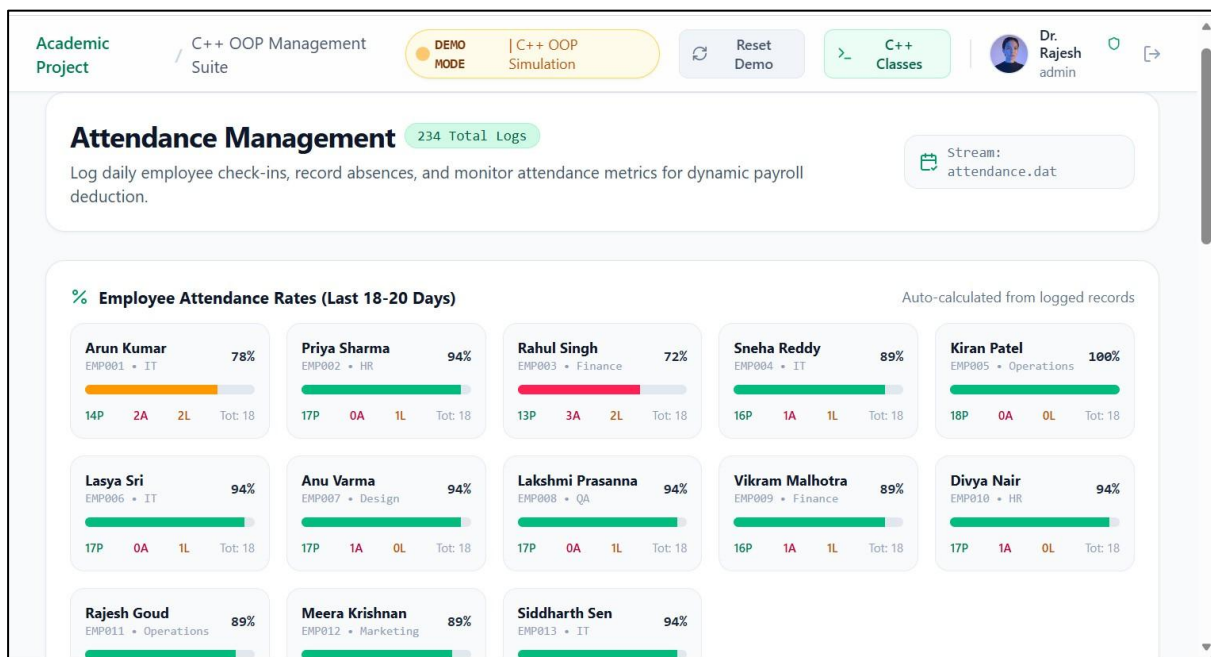
4.8 RESULTS AND DISCUSSION

The proposed system successfully integrates employee information, attendance, payroll, and performance management into a single application.

The attendance module records employee attendance and provides information required for payroll processing. The payroll module automatically calculates salary and attendance-based deductions.

The performance module evaluates employees using performance score, productivity, and attendance and generates performance reports.

The implementation of Classes, Objects, Encapsulation, Inheritance, Polymorphism, and File Handling makes the application modular and maintainable.



Academic Project

C++ OOP Management Suite

DEMO MODE

C++ OOP Simulation

Reset Demo

C++ Classes

Dr. Rajesh admin

Salary Calculation Controller

Select employee to compute live pay slip

Employee Selection *

EMP001 - Arun Kumar (IT • ₹65,000)

Payroll Month

August 2026

Total Working Days

22

Attendance Live Ingestion

Real-time Connected

Present

14

Absent

2

Leave

2

Recalculate Formula

Save & Commit to payroll.dat

View & Print Payslip

Computed Pay Sheet Breakdown

August 2026 • Arun Kumar

Status: Processed

EMPLOYEE ID	DEPARTMENT	PER DAY RATE	ATTENDANCE SCORE
EMP001	IT	₹2,955	64%

ITEM / COMPONENT	FORMULA / LOGIC	AMOUNT (₹)
Basic Salary	Agreed Contract Base	+₹65,000
Allowances (HRA + Travel)	15% of Basic Base	+₹9,750
Absent Deduction	2 days × ₹2955	-₹5,910
Professional Tax	Statutory Flat Deduction	-₹500
Final Net Payable Salary:		₹68,340

C++ Polymorphism Note:

The salary is computed via virtual dispatch: virtual double calculatePayroll() = 0; allowing custom allowances per derived employee type.

Academic Project

C++ OOP Management Suite

DEMO MODE

C++ OOP Simulation

Reset Demo

C++ Classes

Dr. Rajesh admin

Score Entry Form

Weighted formula appraisal

Select Employee *

EMP001 - Arun Kumar (IT • Software Developer)

Productivity & Task Output (35%)

90/100

Attendance & Consistency (25%)

85/100

Deliverable Quality & Accuracy (25%)

90/100

Team Collaboration & Ethics (15%)

88/100

Evaluator Comments & Qualitative Feedback

Strong technical leadership, punctuality, and robust delivery.

Department Performance Scores

Productivity vs Quality vs Overall Score

Q3 Matrix

Employee	Overall Rating	Productivity	Quality
Arun	90	85	95
Priya	85	90	80
Rahul	80	85	75
Kiran	95	90	90
Lasya	90	85	95
Anu	85	90	80
Vikram	80	85	75
Rajesh	75	80	70
Siddharth	90	85	95

Rating Tier Breakdown

5 Staff Appraised

CHAPTER 5

ADVANTAGES, LIMITATIONS AND FUTURE SCOPE

5.1 Advantages

- Reduces manual work.
- Improves payroll accuracy.
- Provides centralized employee information.
- Automates attendance processing.
- Supports employee performance evaluation.
- Reduces data duplication.
- Provides faster information retrieval.
- Generates reports efficiently.
- Demonstrates practical application of C++ OOP concepts.

5.2 Limitations

- The current system uses file handling instead of a database.
- Attendance information depends on the data entered by the administrator.
- The system does not currently implement biometric devices.
- The current version is primarily a C++ academic prototype.
- Advanced web and cloud functionality is not included.

5.3 Future Scope

The system can be enhanced by implementing:

1. Biometric Attendance

Fingerprint or biometric devices can automatically record employee attendance.

2. Face Recognition

Camera-based facial recognition can be used for automated attendance.

3. Database Integration

MySQL or another database can replace file-based storage.

4. Web Application

The system can be converted into a web-based HR management application.

5. Mobile Application

Employees could access attendance, salary and performance information through a mobile application.

6. Cloud Storage

Employee records can be securely stored and accessed through cloud infrastructure.

7. AI-Based Performance Analysis

Machine learning can be incorporated to identify performance trends.

8. Automated Payslip Generation

The system can automatically generate digital salary slips.

CHAPTER 6

CONCLUSION

The Intelligent Employee Attendance, Payroll, and Performance Management System provides an integrated solution for managing important employee-related activities.

The system successfully combines employee information management, attendance tracking, payroll processing, and performance evaluation into a single C++ application.

Advanced Object-Oriented Programming concepts such as Classes, Objects, Encapsulation, Inheritance, and Polymorphism provide a structured approach to system development, while File Handling provides persistent storage for employee records.

The proposed system reduces manual effort, minimizes payroll calculation errors, improves employee-data organization, and provides efficient report generation

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The first five references correspond to the studies listed in your PPT's literature survey/reference section

APPENDIX A – SAMPLE EMPLOYEE DATA

Employee ID	Name	Department	Designation	Salary
101	Employee A	CSE	Developer	₹30,000
102	Employee B	HR	HR Executive	₹35,000
103	Employee C	Finance	Accountant	₹32,000

APPENDIX B – SAMPLE ATTENDANCE DATA

Employee ID	Present	Absent	Leave
101	24	2	1
102	26	1	0
103	22	3	2

APPENDIX C – OOP CONCEPT MAPPING

Project Requirement	C++ Concept
Employee information	Class & Object
Secure employee data	Encapsulation
Different employee categories	Inheritance
Different employee behaviours	Polymorphism
Data storage	File Handling
Employee collection	Arrays/Data Structures
Operations	Member Functions